

COPPER ALLOY ADDITIVE MANUFACTURING FOR SOLEIL II

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Abstract

The Synchrotron SOLEIL is a large-scale research facility in France that provides synchrotron radiation from terahertz to hard X-rays for various scientific applications. To meet the evolving needs of the scientific community and to remain competitive with other European facilities, SOLEIL has planned an upgrade project called SOLEIL II. The project aims to reconstruct the storage ring as a Diffraction Limited Storage Ring (DLSR) with a record low emittance which will enable nanometric resolution.

The mechanical design of this project involves several challenges such as the integration of new magnets, vacuum chambers, insertion devices and beamlines in the existing infrastructure and is mainly based on extensive simulations, prototyping and testing new fabrication methods such as additive manufacturing (AM) to ensure the feasibility, reliability, and performance of several key elements.

This paper presents an overview of the mechanical design R&D and thermomechanical performance verifications on copper alloy parts fabricated in additive manufacturing in the scope of SOLEIL-II project.

INTRODUCTION

Additive manufacturing (AM) is increasingly being considered in the accelerator field to fabricate complex, thermally loaded components such as vacuum absorbers, beam stops, and photon masks. Among the materials under investigation, copper alloys like CuCrZr offer a promising combination of high thermal conductivity and mechanical properties, which are essential for devices exposed to intense synchrotron radiation.

However, components fabricated via AM must be rigorously validated before integration into operational machines, particularly regarding thermal fatigue resistance. In the framework of the SOLEIL II upgrade, a test campaign was launched in collaboration with NSLS-II to study the thermal fatigue lifetime of CuCrZr components manufactured using Direct Metal Laser Sintering (DMLS) method.

SOLEIL II PROJECT

SOLEIL is the French third generation light source operated for users since 2008 with an electron beam emittance of 4 nm·rad at an energy of 2.75 GeV in high intensity (500 mA, multibunch) [1].

The current lattice of the SOLEIL storage ring is composed of 16 modified two-bend achromat cells, 8 of which have short straight sections between the dipoles, altogether

giving a total of 24 straight sections. After years of successful operation, a series of feasibility studies were initiated for an upgrade of the storage ring with a significantly lower emittance.

The SOLEIL Upgrade project, known as SOLEIL-II aims to design and build a 2.75 GeV diffraction-limited synchrotron light source preserving the existing infrastructure, 29 beamlines (far-IR to hard X-rays) and the 500 mA uniform filling pattern. The lattice of the new storage ring presented in CDR report [1] is built over a non-standard combination of twelve 7BA cells and eight 4BA cells [2, 3]. Figure 1 shows the arrangement of the magnets in the 7BA cell and the 4BA cell of SOLEIL-II lattice.

The length of the 7BA cell is rather short (~16 m) containing 52 magnets, depending on the lattice version, and including 7 dipoles. This very high density of multipoles increases the problem of compactness and creates implementation difficulties [4].

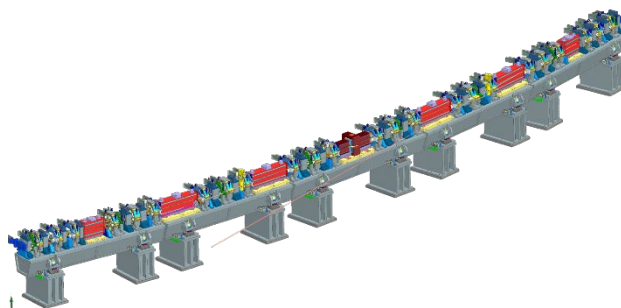


Figure 1: Engineering layout of the 7BA cell type of the new MBA-ARC.

CONTEXT AND MOTIVATION

One of the project's technical challenges is the miniaturization of vacuum chambers (Ø12 mm). Many absorbers and beamline components are expected to undergo high thermal cycling due to their proximity to synchrotron radiation sources like achromat dipoles and insertion devices.

Additive manufacturing (AM) is gaining attention in the accelerator domain for producing complex, thermally stressed components. Among the materials being studied, copper alloys like CuCrZr stand out due to their excellent thermal conductivity and strong mechanical properties, both of which are critical for components operating under intense synchrotron radiation. Additive manufacturing provides the possibility to design and fabricate optimized geometries that would be impossible to achieve using conventional machining [5].

However, the lack of experimental data on thermal fatigue resistance of AM CuCrZr, especially under realistic operating conditions, calls for a dedicated R&D effort. In

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addition, these components must be rigorously validated before integration into operational machines.

As part of SOLEIL II project, a joint test campaign with NSLS-II was conducted to assess the thermal fatigue performance of CuCrZr components fabricated using DMLS process.

BEAM STOP DESIGN

Two beam stops were designed and produced using DMLS procedure by Synchrotron SOLEIL's industrial partner GKN [6]. The hollow in the middle plan of the beam stop aims at thermally decoupling upper and lower hotspots and ensuring uniform stress distribution (Fig. 2). Both samples had the same standard restructuring heat treatment after printing. Afterward, a post-processing involved two different heat treatment protocols has been applied on:

- Sample #1: Subjected to an annealing treatment aimed at maximizing thermal conductivity. The resulting measured value was ~ 350 W/m·K. However, this process led to a reduced hardness of ~ 90 Vickers, barely sufficient for certain vacuum sealing requirements (e.g., knife-edge flanges).
- Sample #2: Treated to increase hardness (~ 150 Vickers) through a hardening heat cycle. The thermal conductivity was slightly lower at ~ 306 W/m·K.

Both samples were machined at the SOLEIL workshop and inspected prior to shipping to BNL site to be installed on the thermal fatigue test bench.

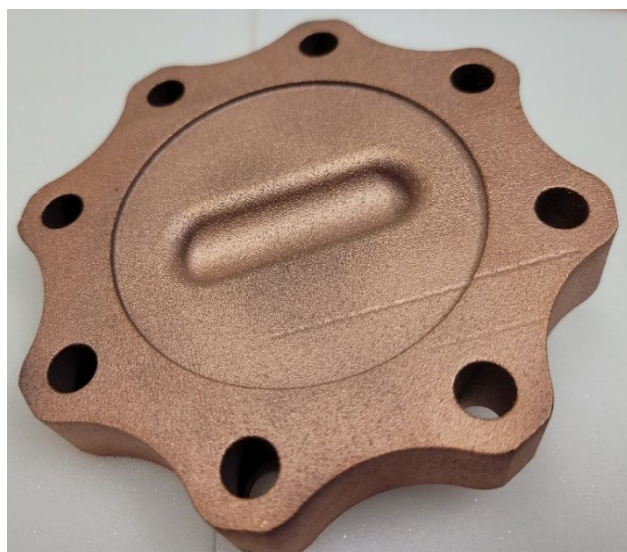


Figure 2: 3D printed beam stop before final machining.

EXPERIMENTAL SETUP

The thermal fatigue tests were carried out at the NSLS-II facility. The setup includes an in-vacuum undulator as the high-power density source, adjustable slits to shape and localize the thermal load on the target and a periodic vertical motion representing one fatigue cycle per oscillation (Fig. 3).

The beam stop is water cooled. The flow rate and the temperature difference between water circuit inlet and outlet are precisely measured in order to calculate the total power absorbed by the cooling system [7].

The samples were tested under ultra-high vacuum conditions, with target temperatures ranging from 250 °C to 360 °C. Each sample underwent 10,000 thermal cycles. Despite minor issues—such as radiation damage to thermocouples—the tests were completed successfully (Fig. 4).

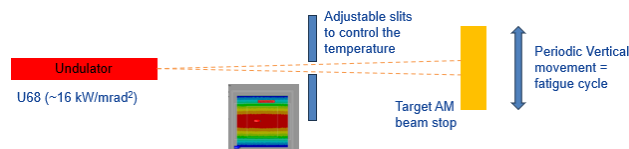


Figure 3: Schematic of NSLS-II (BNL) test setup.

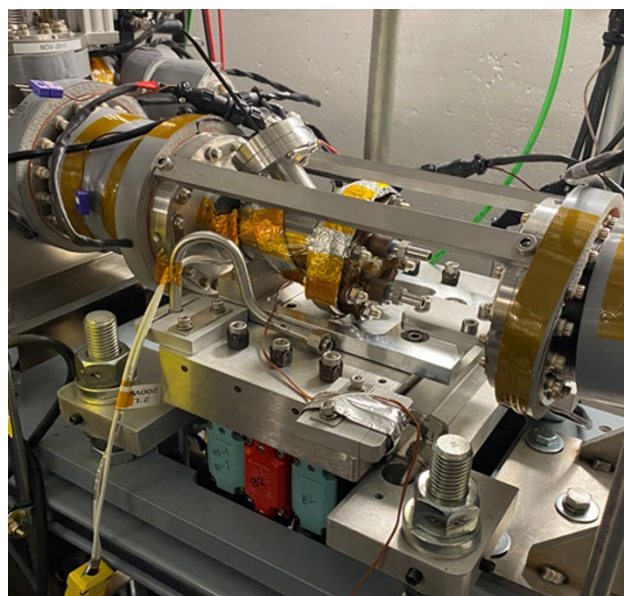


Figure 4: 3D printed beam stop assembled on the vertical movement table ready for fatigue test.

RESULTS AND OBSERVATIONS

Thermal cycling of the beam stops leads to upper and lower beam footprints on the beam stop. Three samples of 10 mm (H) x 15 mm (V) x 10 mm (depth) were cut out of the higher-temperature (360°) beam stop from its two footprints and from an unexposed area approximately 20 mm from the nearest footprint. These samples were sent for microstructural evaluation by L.I.T. lab which provided stereoscopic images, micrographs, and an evaluation report [8]. The images from the top and bottom samples were very similar and only the ones from the top sample are shown below.

Figure 5 (top) shows a stereoscopic image near beam footprint region. It shows machining grooves and some voids on the surface. The Image taken in $135\times$ magnification after polishing (which removes the surface thickness by 50 - 100 μm) shows many voids of different sizes (Fig. 5 bottom).



Figure 5: Stereoscopic (top, 40 X) and after-polishing (bottom, 135 X) images of beam footprint (Side 1).

A micrograph (magnification of X 535) after etching of the top surface (Side 1) shows a fine grain structure and two significant voids of $\sim 25 \mu\text{m}$ size (Fig. 6, top). There appears to be a hairline crack propagating top-left void along the grain boundary. This crack is not of any significant depth as it does not show in the micrograph of the perpendicular-cut surface (Fig. 6, bottom).

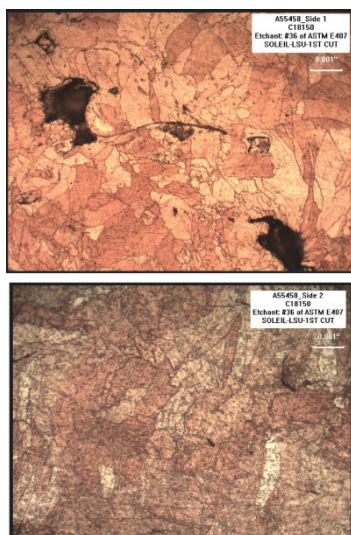


Figure 6: Micrographs (X 535) after etching of the top surface (Side 1) and perpendicular-cut surface (Side 2).

The summary conclusion of the LIT microstructural evaluation is as follows:

- Minor surface damage in the form of "cat-paw" scratches and possible hairline cracks.
- No significant signs of deformation, delamination, or density collapse.

These encouraging results indicate that DMLS-printed CuCrZr can offer thermal fatigue resistance comparable to conventionally manufactured components in bulk. Key findings include:

- Heat treatment is crucial and must be adapted to the specific operational constraints: surface hardness vs. thermal conductivity. A double heat treatment seems necessary: a restructuring annealing followed by a targeted optimization (hardening or annealing depending on the use case).
- The DMLS process achieved densities up to 99.5%, with only minimal impact on performance under tested conditions.

CONCLUSION

This study confirms the potential of 3D printed CuCrZr for demanding synchrotron applications. Beam stops fabricated using DMLS demonstrated robust performance under thermal cycling at high temperatures, with no evidence of fatigue damage up to 10,000 cycles.

While promising, the integration of such components requires careful control of manufacturing parameters, particularly heat treatment strategies. These findings contribute to the growing body of knowledge supporting the adoption of additive manufacturing in accelerator engineering.

ACKNOWLEDGMENTS

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REFERENCES

- [1] Synchrotron SOLEIL Facility website, <https://www.synchrotronsoleil.fr>
- [2] A. Louergue, *et al.*, "TDR Baseline Lattice for the Upgrade of SOLEIL", in *Proc. IPAC'22*, Bangkok, Thailand, Jun. 2022, pp. 1393-1396.
[doi:10.18429/JACoW-IPAC2022-TUPOMS004](https://doi.org/10.18429/JACoW-IPAC2022-TUPOMS004)
- [3] O. R. Blanco-Garcia, *et al.*, "Status of the Soleil Upgrade Lattice Robustness Studies", in *Proc. IPAC'22*, Bangkok, Thailand, Jun. 2022, pp. 433-436.
[doi:10.18429/JACoW-IPAC2022-MOPOTK004](https://doi.org/10.18429/JACoW-IPAC2022-MOPOTK004)
- [4] L. Nadolski, *et al.*, "SOLEIL II project: entrance in the construction phase", presented at IPAC'25, Taipei, Taiwan, Jun. 2025, paper MOPB074 to appear in proceedings.
- [5] K. Tavakoli, *et al.*, "Challenges and Solutions for the Mechanical Design of SOLEIL-II", in *Proc. MEDSI'23*, Beijing, China, Nov. 2023, pp.133-135.
[doi:10.18429/JACoW-MEDSI2023-WEOBM01](https://doi.org/10.18429/JACoW-MEDSI2023-WEOBM01)
- [6] GKN Powder Metallurgy, <https://www.gknpm.com/>
- [7] S. Sharma, *et al.*, "Thermal Fatigue Tests on CuCrZr Photon Shutters", presented at MEDSI'25, Lund, Sweden, Sept. 2025 paper WEP64, this conference.
- [8] L.I.T. Lab, Hauppauge, NY, USA, Certificate No. A55481L, July 25, 2025